Gukhwa Jang · CURRICULUM VITAE

Ph.D., Feb 2023
Urban Studies & Design Lab, Graduate School of Environmental Studies, Seoul National University, Seoul, Republic of Korea
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RESEARCH INTERESTS

Urban microclimate · urban design · shrinking cities · climate gentrification · climate mobility · climate-trapped populations.

EDUCATION

Ph.D. in engineering, Interdisciplinary Program in Landscape Architecture (Landscape Architecture Major) 2019 — 2023
Graduate School of Environmental Studies, Seoul National University
- Dissertation: "Healthy cities: urban transformation and its effects"

M.S., Department of Landscape Architecture (Environmental Landscape Architecture Major) 2015 — 2018
Graduate School of Environmental Studies, Seoul National University
- Thesis: "The study of cyclists' gaze behavior in urban environment"

B.S., Landscape Architecture 2011 — 2015
College of Agriculture and Life Sciences, Kyungpook National University · [Graduation Project] SELFARE

RESEARCH EXPERIENCE

Postdoctoral Researcher Innovative Human Resource Development Education and Research Group for Smart City Global Convergence (BK21 FOUR), Seoul National University 2023 — Present

Visiting Researcher Environmental Planning Institute, Seoul National University 2023

PEER-REVIEWED PUBLICATIONS

APA 7th. Name in bold = author; † = first author. Impact Factor / quartile shown where available. (SCIE / ESCI, then KCI; newest first) Korean-journal (KCI) papers are listed in Korean.

1. **Jang, G.**†, Jeon, Y., You, Y., & Cho, H. R. (2026). What health risks arise in shrinking cities? A systematic review. *Sustainable Cities and Society*, *146*, 107502. https://doi.org/10.1016/j.scs.2026.107502 · SCIE

2. **Jang, G.**†, Kim, S., & Lee, J. S. (2025). Association between urban shrinkage and excess mortality during the COVID-19 pandemic. *Nature Cities*, *2*(8), 708–719. https://doi.org/10.1038/s44284-025-00278-y · ESCI

3. **Jang, G.**†, Kim, S., & Lee, J. S. (2022). Planning scenarios and microclimatic effects: The case of high-density riverside residential districts in Seoul, South Korea. *Building and Environment*, *223*, 109517. https://doi.org/10.1016/j.buildenv.2022.109517 · SCIE

4. **Jang, G.**†, & Kim, S. (2021). Are decline-oriented strategies thermally sustainable in shrinking cities?. *Urban Climate*, *39*, 100924. https://doi.org/10.1016/j.uclim.2021.100924 · SCIE

5. **Jang, G.**†, & Kim, S. (2019). INVESTIGATING THE EFFECT OF A RAISED CYCLE TRACK, PHYSICAL SEPARATION, LAND USE AND NUMBER OF PEDESTRIAN ON CYCLISTS' GAZE BEHAVIOR. *JOURNAL OF ARCHITECTURE AND URBANISM*, *43*(1), 112–122. https://doi.org/10.3846/jau.2019.3786 · ESCI / Scopus

6. **이승현·장국화·송영근·길승호** (2026). **동일 시공간 조건에서의 저층·고층 주거지** LST **산출 방법 비교 - 위성영상, 시뮬레이션**, UAV**를 중심으로** -. ***한국환경복원기술학회지***, *29*(2), 41–59. https://doi.org/10.13087/kosert.2026.29.2.41 · KCI

7. Choi, J., **Jang, G.**, & Kim, D. (2026). Exploring the Changes in the Street Environment under Climate Gentrification. *(Accepted, in press)*.

8. **장국화†** (2026). **한국의 기후 젠트리피케이션 발생 가능성에 대한 탐색적 연구: 거주지 선호, 부동산 가격 영향 인식, 이주의향을 중심으로**. ***게재 확정 (출판 예정)***.

9. **이승현·장국화** (2026). **국내 조경 분야 종사자의 생성형** AI **활용 실태 연구: 과업 단위 활용과 직무군에 따른 차이를 중심으로**. ***게재 확정 (출판 예정)***.

10. **장국화†·김다현** (2025). **기후 젠트리피케이션: 이론, 측정, 동인의 체계적 문헌고찰**. ***국토계획 (대한국토·도시계획학회지)***, *60*(7), 80–91. https://doi.org/10.17208/jkpa.2025.12.60.7.80 · KCI

MANUSCRIPTS UNDER REVIEW

APA 7th. Name in bold = author; † = first author. (10 manuscripts; first-author listed first)

1. Kang, S., **Jang, G.**, & Kim, D. ***산업단지 열환경 분석을 위한 산업특화형 지역기후구역****(ILCZ)* ***정의와 유형화*** [In revision].

2. Yoo, Y., Jeon, Y., Cho, H., & **Jang, G.** *How Do Shrinking Cities Restructure? A Three-Zone Analysis of Spatial Differentiation in South Korea* [In preparation].

3. **Jang, G.†**, Ha, J., & Cho, H. *Do superblocks ensure thermal comfort? Planning benefits and microclimatic trade-offs of a pedestrian-oriented urban form* [In preparation].

4. Cho, H., **Jang, G.**, Yoo, Y., & Jeon, Y. *The morphological paradox of fine-grained urban form: Resilience in shrinking neighbourhoods* [In revision].

5. Jeon, Y., Yoo, Y., Cho, H., & **Jang, G.** *Spatiotemporal dynamics of population in South Korea's medium-sized cities: Expanding the model of urban development stages from an urban shrinkage perspective* [Submitted to SCIE journal].

6. Lee, J., **Jang, G.**, & Kim, D. *Unequal Displacement in the Climate Crisis: Developing Climate Gentrification Risk Indicators Using Delphi and AHP* [Submitted to SCIE journal].

7. Hong, H., **Jang, G.**, & Kim, D. *Climate-trapped populations: What drives households to remain in high flood-risk housing in South Korea?* [In preparation].

8. Oh, J., **Jang, G.**, & Kim, D. *Post-disaster gentrification in the wake of Hurricane Harvey: Flood exposure and neighborhood change in Harris County* [Submitted to SCIE journal].

9. **Jang, G.†** *Beyond Aspiration and Capability: Necessity as a Gate to Climate (Im)mobility in South Korea* [In preparation].

10. Koo, S., **Jang, G.**, & Kim, D. *Does Climate Gentrification Occur in Korea?: A Municipal Panel Analysis of the Effects of Flood, Earthquake, and Heat Exposure on Land Prices and Residential Composition* [In preparation].

BOOKS

1. (2021). *Smart Land Management and Planning Interventions: Learning from South Korea's Half-century Experiences*. Seoul National University Press. ISBN 9788952130976.

2. (2021). *New Normal City*. Urban Issue. ISBN 9791197343100.

FELLOWSHIPS & FUNDING

Competitive fellowships / scholarships (amounts converted at ₩1,400 = $1).

기후 젠트리피케이션: 기후 불평등의 공간화를 통해 바라본 기후불평등과 인권 위기, National Research Foundation of Korea (NRF) (PI) 2025 — 2027

노후 산업단지의 열적 쾌적성 개선을 위한 공간 모델 개발, National Research Foundation of Korea (NRF) (PI) 2024 — 2027

쇠퇴도시 내 유휴공터를 활용한 기후 위기 대응형 공간 모델 개발, National Research Foundation of Korea (NRF) (PI) 2023 — 2024

Fund for Short-term visiting scholar, Korea Foundation for Women In Science, Engineering and Technology (WISET) 2024 — 2024

Scholarship, Brain Korea 21 (BK21) FOUR 2020 — 2023

Scholarship, 3rd Brain Korea 21 (BK21) 2019 — 2020

Scholarship for academic excellence, SNU GSES 2016

University Scholarship for Academic Excellence and Achievement, Seoul National University 2016

University Scholarship for Academic Excellence and Achievement, Kyungpook National University 2011

INVITED TALKS

- **Gukhwa Jang** (2025). Lecture in **자원순환형 도시설계 연구**. *Guest lecture, Seoul National University.*
- **Gukhwa Jang** (2025). Advanced Urban Environment Analysis and Data Visualization. *Guest lecture, Seoul National University.*
- **Gukhwa Jang** (2024). Microclimate Simulation (ENVI-met). *Invited seminar, Gyeonggi Research Institute, Suwon, South Korea.*

CONFERENCE PRESENTATIONS (INTERNATIONAL)

First author / presenter unless marked "team". 10 domestic presentations (KOSERT, KSEE, KILA) available on request.

- (2026). Climate-trapped populations in seoul's flood-prone semi-basement housing. *[Poster] The 62nd IFLA World Congress, Hong Kong.*
- (2026). How does climate risk reshape neighborhoods? climate gentrification. *[Poster] The World Planning Schools Congress (WPSC), Helsinki, Finland.*
- (2025). A new framework for urban heat island management: assessing the impact of industrial complex-microclimate zones (ic-mcz) on urban heat islands. *[Poster] AGU 25, New Orleans, USA.*
- (2025). Strategies for managing vacant lots and enhancing thermal comfort in shrinking cities. *[Oral] The International Making Cities Livable (IMCL), Potsdam, Germany.*
- (2024). Assessing the applicability of UAV thermal drones for urban microclimate evaluation. *[Poster] Ecological Society of America (ESA), Long Beach, USA.*
- (2024). The health inequality in shrinking cities: the effects of urban shrinkage on excess deaths during the covid-19 pandemic. *[Poster] AESOP, Paris, France.*
- (2024). Urban green infrastructure planning for urban microclimate. *[Oral] NEOCITY EPOWERMENT FORUM 2024 (NEF), Florida, USA.*
- (2023). The challenge of shrinking cities: a cross-city comparative study. *[Oral] The 59th International Making Cities Livable (IMCL), Dorchester, UK.*
- (2022). The evaluation of heat-mitigation strategies on outdoor heat stress in the waterfront public spaces. *[Oral] The 29th International Seminar on Urban Form (ISUF), Online.*
- (2021). Investigating the effects of urban structures on urban heat stress in shrinking cities. *[Oral] The 28th International Seminar on Urban Form (ISUF), Online.*

- (2020). How do different planning approaches affect heat stress in shrinking cities?. *[Oral] The 14th International Association for China Planning (IACP), Online.*

RESEARCH PROJECTS

National R&D and municipal projects (newest first). Budget = total project funding (national R&D).

Climate Gentrification: Climate Inequality and the Human Rights Crisis
2025 — Present · *Seoul National University* · ***Role: Principal Investigator*** · *PI: Gukhwa Jang*

폭염 취약지역 지원을 위한 기상사회경제 융합 폭염정보 개발
2025 · *Gwangju Regional Office of Meteorology* · ***Role: Research Assistant***

Development of a Spatial Model for Improving Thermal Comfort in Aging Industrial Complexes
2024 — Present · *Seoul National University* · ***Role: Principal Investigator*** · *PI: Gukhwa Jang*

Development of climate change response spatial model utilizing vacant lots in shrinking cities
2023 — 2024 · *Seoul National University* · ***Role: Principal Investigator*** · *PI: Gukhwa Jang*

디지털트윈 시뮬레이션 기술 기반 스마트시티 통합 설계 솔루션 개발
2021 — 2023 · ***Role: Researcher***

Palliative planning and shrinking-smart model for declined areas associated with a super-aged, low fertility society
2020 — 2023 · ***Role: Researcher*** · *PI: Saehoon Kim*

Inception Report for Utilizing Korean Support and Partnerships for Green Growth in FCV
2020 · ***Role: Researcher*** · *PI: Saehoon Kim*

BK21 3단계, 4단계 플러스사업
2019 — 2023 · ***Role: Researcher***

노인의료복지시설 화재안전 실태조사 및 개선방안 연구
2018 · ***Role: Research Assistant***

지속가능한 따복공동체 마을공유공간 활성화 방안 연구
2016 · ***Role: Principal Investigator*** · *PI: Gukhwa Jang*

야외전시장 재배치 및 다목적 컨벤션센터 기본 개념 수립
2016 · ***Role: Research Assistant***

안산시 고잔1동 오픈스페이스 개선 방안에 관한 연구
2015 · ***Role: Researcher***

쇠퇴하는 저층주거 에너지 빈공층의 환경개선과 복합용도개발 유도 모델
2015 — 2017 · ***Role: Research Assistant***

PATENTS (REPUBLIC OF KOREA)

3 registered, 0 pending.

Registered

- **수목 관리 플랫폼 제공 시스템과 방법 및 이를 위한 컴퓨터 프로그램** *Registered Sep 2025* · ***이승현, 장국화***
- **이미지 기반의 가로수 정보 관리 시스템과 방법 및 이를 위한 컴퓨터 프로그램** *Registered Nov 2024* · ***이승현***
- History marker *Registered Nov 2021* · ***서울대학교산학협력단***

Pending

AWARDS & HONORS

Awards

Prize, Data Visualization and Idea Contest on Public Safety in Suwon City — LH COMPAS 2021

2020

Prize, The 13th National Exhibition of Korean Landscape Architecture — The Korean Institute of Landscape Architecture 2016

Prize for Excellence, The 11th National Exhibition of Korean Landscape Architecture — The Korean Institute of Landscape Architecture 2014

Prize for Excellence, Capstone Design Contest — Kyungpook National University 2014

Grand Prize, Graduation Exhibition — Kyungpook National University 2014

Prize, The 10th National Exhibition of Korean Landscape Architecture — The Korean Institute of Landscape Architecture 2013

TEACHING EXPERIENCE

6 university courses · 6 invited lectures and workshops.

University Courses

- Research Methods in Urban Landscape Architecture (3 credits), Sungkyunkwan University
  Co-instructor: Myungjin Shin, Ph.D. Taught quantitative research methods in landscape architecture, with a focus on research design, variable specification, data collection, and statistical analysis for Master's and Ph.D. students
  Spring 2026
- Smart City Seminar on Industrial Technology (Smart Urban Planning in Response to Climate Change) (3 credits), Seoul National University
  Taught the fundamentals of urban microclimate and applied spatial analysis through QGIS and ENVI-met simulations for graduate students (Master's and Ph.D. levels) Spring 2025
- Understanding and Application of Spatial Information (3 credits), Incheon National University
  Taught urban data interpretation and spatial analysis using QGIS to support urban research Fall 2023
- Survey and Data Analysis for Social Science 2 (3 credits), The University of Suwon
  Taught basic statistical concepts and applied analysis using R for social science research, designed for second-year undergraduate students in urban real estate studies Fall 2023
- Survey and Data Analysis for Social Science 1 (3 credits), The University of Suwon
  Taught the fundamentals of social science data analysis and hands-on exercises using R for second-year undergraduate students in urban real estate studies Spring 2023
- Statistics (3 credits), The University of Suwon
  Taught fundamental statistical principles for first-year students in urban real estate Spring 2023

Invited Lectures & Workshops (6 sessions, 2024 — 2025)

- Microclimate Simulation (ENVI-met), Pusan National University Dec 2025
- Microclimate Simulation (ENVI-met), Pusan National University Jun 2025
- Microclimate Simulation (ENVI-met) day 1–3, Korea University Apr 2024

PROFESSIONAL EXPERIENCE

Lecturer Sungkyunkwan University 2026

Lecturer College of Engineering, Seoul National University 2025 — Present

Lecturer Division of Architecture & Urban Design, Incheon National University 2023 — 2024

Visiting Professor Division of Architecture for Urban Planning & Real Estate Development, College of Engineering, University of Suwon 2023 — 2024

REFERENCES

Available upon request.

TIMELINE (2023 — PRESENT)

All confirmed activity since January 2023 in a single chronology (newest first). Manuscripts under review are not included.

2026

Oct CONFERENCE — Climate-trapped populations in seoul's flood-prone semi-basement housing · *The 62nd IFLA World Congress, Hong Kong*

Aug PUBLICATION — What health risks arise in shrinking cities? A systematic review · *Sustainable Cities and Society*

Jun CONFERENCE — How does climate risk reshape neighborhoods? climate gentrification · *The World Planning Schools Congress (WPSC), Helsinki, Finland*

Apr PUBLICATION — **동일 시공간 조건에서의 저층·고층 주거지** LST **산출 방법 비교 - 위성영상**, **시뮬레이션**, UAV**를 중심으로** - · ***한국환경복원기술학회지***

Mar TEACHING — Research Methods in Urban Landscape Architecture (3 credits), Sungkyunkwan University · *Spring 2026*

Mar POSITION — Lecturer · *Sungkyunkwan University*

PUBLICATION — Exploring the Changes in the Street Environment under Climate Gentrification · *(Accepted, in press)*

PUBLICATION — **국내 조경 분야 종사자의 생성형** AI **활용 실태 연구**: **과업 단위 활용과 직무군에 따른 차이를 중심으로** · ***게재 확정 (출판 예정)***

PUBLICATION — **한국의 기후 젠트리피케이션 발생 가능성에 대한 탐색적 연구**: **거주지 선호**, **부동산 가격 영향 인식**, **이주의향을 중심으로** · ***게재 확정 (출판 예정)***

2025

Dec PUBLICATION — **기후 젠트리피케이션**: **이론**, **측정**, **동인의 체계적 문헌고찰** · ***국토계획 (대한국토·도시계획학회지)***

Dec CONFERENCE — A new framework for urban heat island management: assessing the impact of industrial complex-microclimate zones (ic-mcz) on urban heat islands · *AGU 25, New Orleans, USA*

Dec TEACHING — Microclimate Simulation (ENVI-met), Pusan National University

Nov INVITED TALK — Lecture in **자원순환형 도시설계 연구** · *Guest lecture, Seoul National University*

Oct CONFERENCE — Strategies for managing vacant lots and enhancing thermal comfort in shrinking cities · *The International Making Cities Livable (IMCL), Potsdam, Germany*

Sep INVITED TALK — Advanced Urban Environment Analysis and Data Visualization · *Guest lecture, Seoul National University*

Sep PATENT — **수목 관리 플랫폼 제공 시스템과 방법 및 이를 위한 컴퓨터 프로그램** · *Registered* · ***이승현, 장국화***

Jul PUBLICATION — Association between urban shrinkage and excess mortality during the COVID-19 pandemic · *Nature Cities*

Jun TEACHING — Microclimate Simulation (ENVI-met), Pusan National University

Jun PROJECT — Climate Gentrification: Climate Inequality and the Human Rights Crisis (PI) · *Seoul National University*

Jun FUNDING — **기후 젠트리피케이션**: **기후 불평등의 공간화를 통해 바라본 기후불평등과 인권 위기**, National Research Foundation of Korea (NRF) (PI)

Mar TEACHING — Smart City Seminar on Industrial Technology (Smart Urban Planning in Response to Climate Change) (3 credits), Seoul National University · *Spring 2025*

Mar PROJECT **폭염 취약지역 지원을 위한 기상사회경제 융합 폭염정보 개발** · *Gwangju Regional Office of Meteorology*

Mar POSITION Lecturer · *College of Engineering, Seoul National University*

2024

Nov PATENT **이미지 기반의 가로수 정보 관리 시스템과 방법 및 이를 위한 컴퓨터 프로그램** · *Registered* · ***이승현***

Sep PROJECT Development of a Spatial Model for Improving Thermal Comfort in Aging Industrial Complexes (PI) · *Seoul National University*

Sep FUNDING **노후 산업단지의 열적 쾌적성 개선을 위한 공간 모델 개발**, National Research Foundation of Korea (NRF) (PI)

Aug CONFERENCE Assessing the applicability of UAV thermal drones for urban microclimate evaluation · *Ecological Society of America (ESA), Long Beach, USA*

Aug INVITED TALK Microclimate Simulation (ENVI-met) · *Invited seminar, Gyeonggi Research Institute, Suwon, South Korea*

Jul CONFERENCE The health inequality in shrinking cities: the effects of urban shrinkage on excess deaths during the covid-19 pandemic · *AESOP, Paris, France*

Jul FUNDING Fund for Short-term visiting scholar, Korea Foundation for Women In Science, Engineering and Technology (WISET)

Apr TEACHING Microclimate Simulation (ENVI-met) day 1–3, Korea University

Jan CONFERENCE Urban green infrastructure planning for urban microclimate · *NEOCITY EPOWERMENT FORUM 2024 (NEF), Florida, USA*

2023

Oct CONFERENCE The challenge of shrinking cities: a cross-city comparative study · *The 59th International Making Cities Livable (IMCL), Dorchester, UK*

Sep TEACHING Survey and Data Analysis for Social Science 2 (3 credits), The University of Suwon · *Fall 2023*

Sep TEACHING Understanding and Application of Spatial Information (3 credits), Incheon National University · *Fall 2023*

Sep PROJECT Development of climate change response spatial model utilizing vacant lots in shrinking cities (PI) · *Seoul National University*

Sep FUNDING **쇠퇴도시 내 유휴공터를 활용한 기후 위기 대응형 공간 모델 개발**, National Research Foundation of Korea (NRF) (PI)

Sep POSITION Lecturer · *Division of Architecture & Urban Design, Incheon National University*

Aug POSITION Postdoctoral Researcher · *Innovative Human Resource Development Education and Research Group for Smart City Global Convergence (BK21 FOUR), Seoul National University*

Apr POSITION Visiting Researcher · *Environmental Planning Institute, Seoul National University*

Mar TEACHING Statistics (3 credits), The University of Suwon · *Spring 2023*

Mar TEACHING Survey and Data Analysis for Social Science 1 (3 credits), The University of Suwon · *Spring 2023*

Mar POSITION Visiting Professor · *Division of Architecture for Urban Planning & Real Estate Development, College of Engineering, University of Suwon*

Last updated: Nov 2026